Start with the starting position

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Prediction range / ES: intervalo de predicción / PT: intervalo de previsão

A ± bound. “10 ± 2 m” means 8 m to 12 m. More time usually means a wider range.

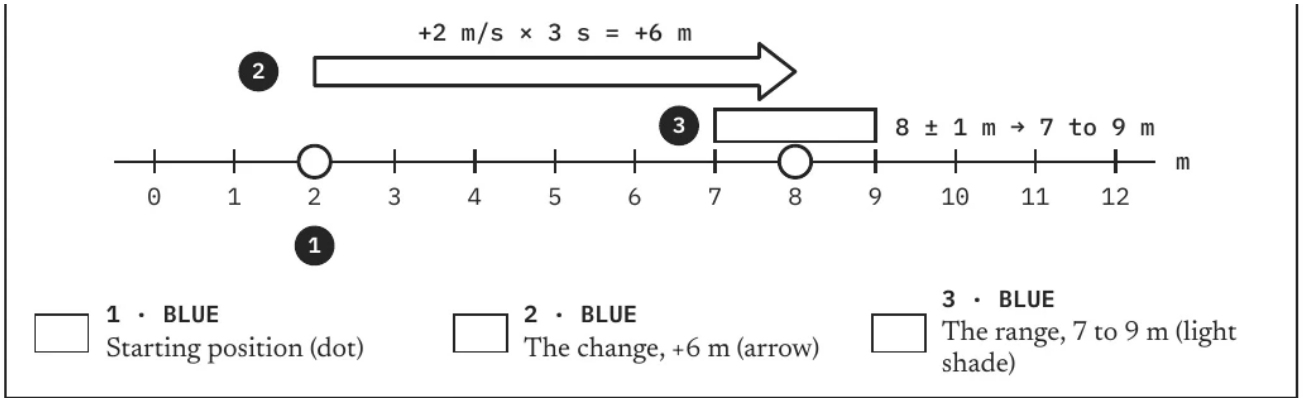
Also on your note page: Starting position, Displacement, Final position.

B READ THE RELATIONSHIP

A forecast gives a position at a future time. A range shows which positions the model allows.

Final position = starting position + velocity × time

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A cart starts at 8 m and moves at –2 m/s for 3 s. Where is it at the end?

Given $x_i = 8$ m $v = -2$ m/s $t = 3$ s

Rule $x_f = x_i + v \times t$

Work change = $(-2)(3) = -6$ m $x_f = 8 - 6$

Answer 2 m moved 6 m left

E SAVE FOR THE FINAL MEETING

With a starting position and a velocity, we can forecast where XJ will be. But every measurement has error, so a responsible forecast is a range, not a single point.

Read the physics: Prediction range

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Start from the actual starting position. At steady velocity, multiply velocity by time to get the displacement. Add that signed change to the starting position.

02 HOW IT WORKS

Measurements have limits. Mark the central forecast, then the smallest and largest allowed positions. With exact time, allow for the starting-position error and the velocity error multiplied by time.

03 WHAT TO CHECK

Use compatible units before multiplying. A range gives limits; it does not tell you the chance of landing at one particular point.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart starts at 8 m and moves at −2 m/s for 3 s. Where is it at the end? 2 m moved 6 m left.

05 CONNECT TO XJ

With a starting position and a velocity, we can forecast where XJ will be. But every measurement has error, so a responsible forecast is a range, not a single point.

SAY IT BEFORE YOU WRITE

What can make the range wider? My center is ___. The allowed positions run from ___ to ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Starting position

ES: posición inicial / PT: posição inicial

Meaning in my words	How can you tell?
How would you explain starting position to a friend? In my words, this means ___.	At 0 s the cart is beside the +5 m mark. Where does its position graph begin? I can tell because ___.
Example or sketch	**Non-example and why**
The cart starts at +5 m and moves 2 m left. What two positions would you mark? My example shows ___.	"The clock starts at zero, so the cart starts at zero." What do the two zeros mean? I would change ___ because ___.

Displacement

ES: desplazamiento / PT: deslocamento

Meaning in my words	How can you tell?
How would you explain displacement to a friend? In my words, this means ___.	Which two marks tell you the displacement even if the cart turns around? I can tell because ___.
Example or sketch	**Non-example and why**
Go from 2 m to 6 m and back to 4 m. What single arrow shows the displacement? My example shows ___.	Someone adds both legs and gets 6 m. What did they find? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Final position

ES: posición final / PT: posição final

Meaning in my words	How can you tell?
How would you explain final position to a friend? In my words, this means ___.	What must you combine with the starting position to locate the finish? I can tell because ___.
Example or sketch	**Non-example and why**
A cart starts at +5 m and moves 7 m left. Where would you draw it? My example shows ___.	A partner reports 7 m because that is how far the cart traveled. What is missing? I would change ___ because ___.

Prediction range

ES: intervalo de predicción / PT: intervalo de previsão

Meaning in my words	How can you tell?
How would you explain prediction range to a friend? In my words, this means ___.	A forecast reads 10 ± 2 km. Which positions are at its ends? I can tell because ___.
Example or sketch	**Non-example and why**
Where would you shade the allowed positions on a number line? My example shows ___.	Someone circles only 10 km. What information disappears? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Mark the central forecast and shade its interval. Label one reason the interval could grow.

Labels you can use: center · lower end · upper end.

Choose zero, direction, and scale.



Tell a partner

My center is ___. The allowed positions run from ___ to ___.

How wide should we search?

05

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Read forecast bounds without inventing a probability.

CHOOSE ONE JOB TOGETHER

SURVEY ENGINEER ☐

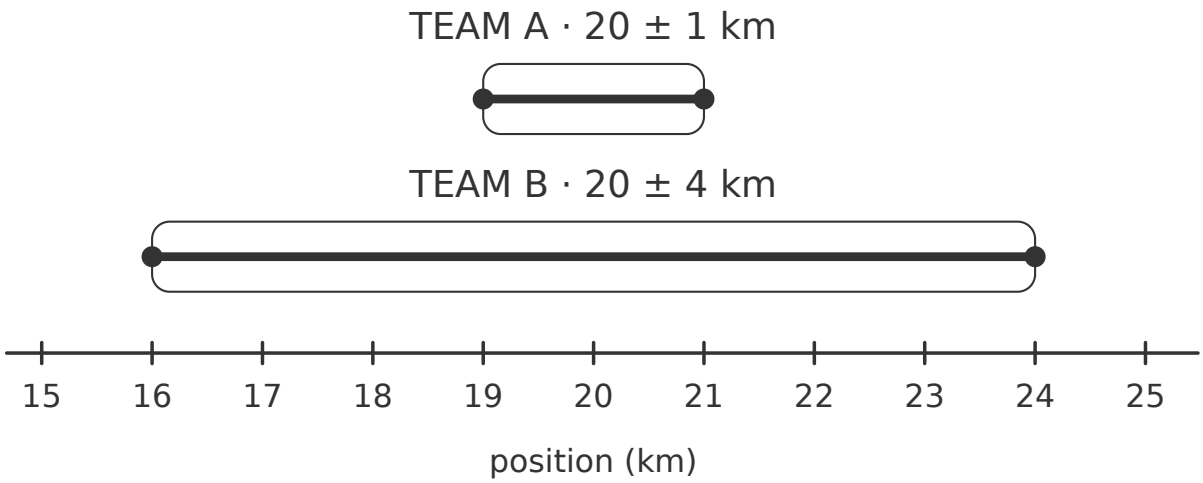
Label both search bands with their endpoints. Recommend which report needs a wider telescope search.

WEATHER-DESK PRESENTER ☐

Present both forecast bands. Explain which search is wider and what the bands do not tell us about impact.

Word help: center / range / endpoints

Two XJ teams predict a center of **20 km**. Team A gives **±1 km**; team B gives **±4 km**. A telescope has one narrow field.



01 OUR FIRST CALL

What search area should the crew prepare for? Use both sets of endpoints.

Team ___ needs a wider search because ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the editor's new label. What can you report from these bands?

Now we would say ___ because ___.

Switch speakers. Does a wider band tell us the chance of impact?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks